

中国海洋大学全日制本科课程期末考试试卷

2026 年 Spring 季学期 考试科目: Statistical Modelling and Inference

学院: 海德学院

试卷类型: SAMPLE EXAM 1 SOLUTIONS 命题人: Thomas Shang 课程教师:
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Examination instructions: This is a closed-book sample examination. A non-programmable calculator is permitted. Show all working. Formulae and statistical quantities required for each question are provided in the question.

Question	1	2	3	4	5	Total
Marks	20	20	20	20	20	100

Name: _____ Student ID: _____

Major/Year: _____ Seat Number: _____

Examination Room: _____ Instructor: _____

Sample Examination 1: Reference Solutions

Five questions, 20 marks each

End of cover page. Begin Question 1 on the next page.

Question 1 [20 marks total]

Bias, variance, MSE, and constructing an unbiased estimator

Let $Y_1, \dots, Y_n$ be a random sample from a population with mean μ and variance σ^2 . Consider the estimator

$$T = \frac{n-1}{n} \bar{Y}, \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

For an estimator $\hat{\theta}$ of θ :

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta, \quad \text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

Also $E(\bar{Y}) = \mu$ and $\text{Var}(\bar{Y}) = \sigma^2/n$.

- (a) Find the bias of T as an estimator of μ . [5 marks]
- (b) Find $\text{Var}(T)$. [5 marks]
- (c) Find $\text{MSE}(T)$. [5 marks]
- (d) Define $T^* = aT$. Find a such that T^* is unbiased for μ . [5 marks]

Solution

- (a) Since $E(T) = \frac{n-1}{n} E(\bar{Y}) = \frac{n-1}{n} \mu$,

$$\text{Bias}(T) = \frac{n-1}{n} \mu - \mu = -\frac{\mu}{n}.$$

- (b)

$$\text{Var}(T) = \left(\frac{n-1}{n} \right)^2 \text{Var}(\bar{Y}) = \frac{(n-1)^2}{n^2} \frac{\sigma^2}{n} = \frac{(n-1)^2 \sigma^2}{n^3}.$$

- (c)

$$\text{MSE}(T) = \frac{(n-1)^2 \sigma^2}{n^3} + \frac{\mu^2}{n^2}.$$

- (d) We need $E(aT) = \mu$, so

$$a \frac{n-1}{n} \mu = \mu.$$

Hence $a = n/(n-1)$ and $T^* = \bar{Y}$.

Question 2 [20 marks total]

Confidence interval and hypothesis test for a population mean

A supermarket manager records the shopping times of $n = 64$ randomly selected customers. The sample mean is $\bar{y} = 33$ minutes. The population standard deviation is known to be $\sigma = 16$ minutes.

For known σ ,

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}.$$

Use $z_{0.025} = 1.96$, $z_{0.05} = 1.645$, and $\Phi(1.50) = 0.9332$.

- (a) Construct a 95% confidence interval for μ . [6 marks]
- (b) Test $H_0 : \mu = 30$ against $H_a : \mu > 30$ at $\alpha = 0.05$. State the test statistic and conclusion. [7 marks]
- (c) Compute the one-sided p-value using the provided normal table value. [4 marks]
- (d) Explain the practical conclusion in one sentence. [3 marks]

Solution

The standard error is

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{16}{8} = 2.$$

(a)

$$33 \pm 1.96(2) = 33 \pm 3.92 = (29.08, 36.92).$$

(b)

$$Z_{\text{obs}} = \frac{33 - 30}{2} = 1.50.$$

For a right-tailed test at $\alpha = 0.05$, reject if $Z_{\text{obs}} \geq 1.645$. Since $1.50 < 1.645$, fail to reject H_0 .

(c)

$$p = P(Z \geq 1.50) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668.$$

- (d) The data do not give sufficient evidence at the 5% level that the true mean shopping time exceeds 30 minutes.

Question 3 [20 marks total]

Multiple linear regression and least squares

Five houses are observed. Price is measured in thousands of AUD, age in decades, and area in thousand square metres.

i	1	2	3	4	5
y_i	45	52	61	68	74
x_{1i} (age)	1	2	3	4	5
x_{2i} (area)	2	1	3	4	5

Fit the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$.

The least-squares normal equations are

$$(X^\top X)\hat{\beta} = X^\top y.$$

For this data set,

$$X^\top X = \begin{pmatrix} 5 & 15 & 15 \\ 15 & 55 & 56 \\ 15 & 56 & 55 \end{pmatrix}, \quad X^\top y = \begin{pmatrix} 300 \\ 978 \\ 957 \end{pmatrix}.$$

- (a) Write down the design matrix X and vector y . [4 marks]
- (b) Solve the normal equations for $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. [10 marks]
- (c) Give the fitted regression equation. [3 marks]
- (d) Predict the price for a house with age $x_1 = 3$ and area $x_2 = 4$. [3 marks]

Solution

(a)

$$X = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 3 \\ 1 & 4 & 4 \\ 1 & 5 & 5 \end{pmatrix}, \quad y = \begin{pmatrix} 45 \\ 52 \\ 61 \\ 68 \\ 74 \end{pmatrix}.$$

(b) The equations are

$$5b_0 + 15b_1 + 15b_2 = 300,$$

$$15b_0 + 55b_1 + 56b_2 = 978,$$

$$15b_0 + 56b_1 + 55b_2 = 957.$$

Subtracting the third equation from the second gives $-b_1 + b_2 = 21$, so $b_2 = b_1 + 21$.

Solving with the first two equations gives

$$\hat{\beta}_0 = 36.6, \quad \hat{\beta}_1 = -6.3, \quad \hat{\beta}_2 = 14.7.$$

(c)

$$\hat{y} = 36.6 - 6.3x_1 + 14.7x_2.$$

(d)

$$\hat{y} = 36.6 - 6.3(3) + 14.7(4) = 76.5.$$

The predicted price is 76.5 thousand AUD.

Question 4 [20 marks total]

Parameter estimation in simple linear regression

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where X_i are fixed, $E(Y_i) = \beta_0 + \beta_1 X_i$, $\text{Var}(Y_i) = \sigma^2$, and the Y_i are independent. Define

$$S_{xx} = \sum_{i=1}^n (X_i - \bar{X})^2, \quad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Use

$$\hat{\beta}_1 = \sum_{i=1}^n a_i Y_i, \quad a_i = \frac{X_i - \bar{X}}{S_{xx}}, \quad \sum a_i = 0, \quad \sum a_i X_i = 1.$$

- (a) Prove that $E(\hat{\beta}_1) = \beta_1$. [6 marks]
- (b) Prove that $\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$. [5 marks]
- (c) Prove that $E(\hat{\beta}_0) = \beta_0$. [5 marks]
- (d) State the covariance $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$ and explain its sign when $\bar{X} > 0$. [4 marks]

Solution

(a)

$$E(\hat{\beta}_1) = \sum a_i E(Y_i) = \sum a_i (\beta_0 + \beta_1 X_i) = \beta_0 \sum a_i + \beta_1 \sum a_i X_i = \beta_1.$$

(b) By independence,

$$\text{Var}(\hat{\beta}_1) = \sum a_i^2 \sigma^2 = \sigma^2 \frac{\sum (X_i - \bar{X})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

(c)

$$E(\bar{Y}) = \frac{1}{n} \sum (\beta_0 + \beta_1 X_i) = \beta_0 + \beta_1 \bar{X}.$$

Hence

$$E(\hat{\beta}_0) = E(\bar{Y}) - \bar{X} E(\hat{\beta}_1) = \beta_0 + \beta_1 \bar{X} - \bar{X} \beta_1 = \beta_0.$$

(d)

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}.$$

If $\bar{X} > 0$, this is negative: when the fitted slope is higher, the fitted intercept tends to be lower because the fitted line passes through $(\bar{X}, \bar{Y})$.

Question 5 [20 marks total]

Binomial MLE and Fisher information

Let $Y \sim \text{Binomial}(n, p)$. In one study, $n = 50$ trials give $y = 18$ successes.

The probability mass function is

$$f(y; p) = \binom{n}{y} p^y (1 - p)^{n-y}.$$

For a scalar parameter p ,

$$I(p) = -E \left[\frac{\partial^2 \ell(p)}{\partial p^2} \right].$$

- (a) Write the likelihood and log-likelihood for p . [4 marks]
- (b) Derive the score equation and find the MLE $\hat{p}$. [6 marks]
- (c) Calculate the numerical MLE for the data. [3 marks]
- (d) Derive the Fisher information $I(p)$ for one binomial observation with fixed n . [5 marks]
- (e) Estimate the approximate standard error of $\hat{p}$ using $I(\hat{p})^{-1/2}$. [2 marks]

Solution

(a)

$$L(p) = \binom{n}{y} p^y (1 - p)^{n-y}, \quad \ell(p) = \log \binom{n}{y} + y \log p + (n - y) \log(1 - p).$$

(b)

$$\ell'(p) = \frac{y}{p} - \frac{n - y}{1 - p}.$$

Setting this to zero gives $y(1 - p) = (n - y)p$, so $y = np$ and

$$\hat{p} = \frac{y}{n}.$$

(c) $\hat{p} = 18/50 = 0.36$.

(d)

$$\ell''(p) = -\frac{Y}{p^2} - \frac{n - Y}{(1 - p)^2}.$$

Since $E(Y) = np$,

$$I(p) = \frac{np}{p^2} + \frac{n - np}{(1 - p)^2} = \frac{n}{p} + \frac{n}{1 - p} = \frac{n}{p(1 - p)}.$$

(e)

$$\text{se}(\hat{p}) \approx I(\hat{p})^{-1/2} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.36(0.64)}{50}} \approx 0.0679.$$

End of examination questions.

Check that you have attempted all five questions.